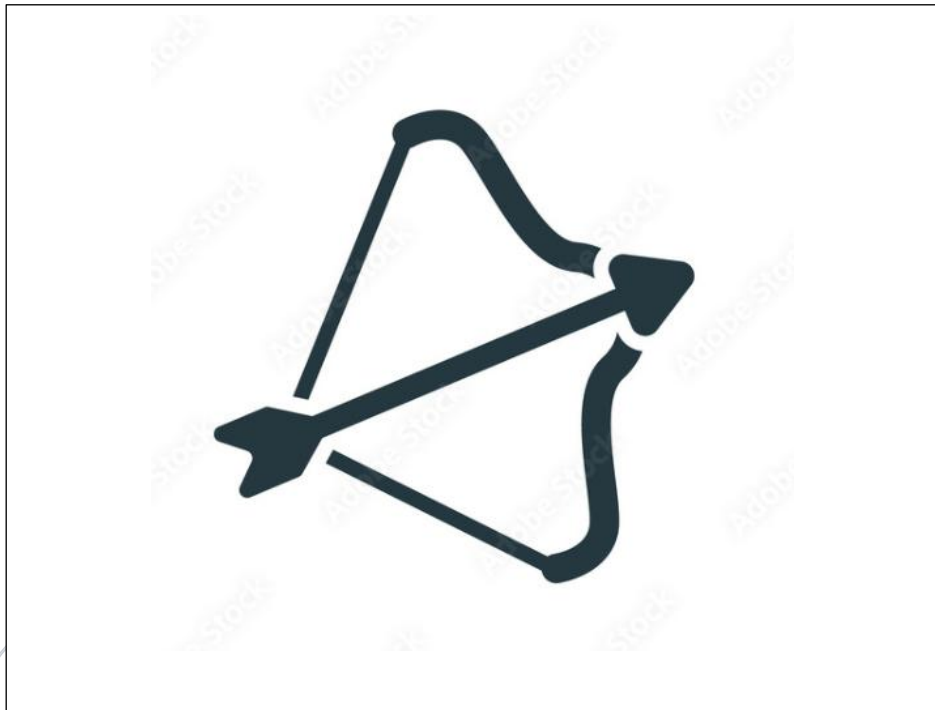


WORLD ROBOT OLYMPIAD FUTURE ENGINEERS CHALLENGE

ENGINEERING
DOCUMENTATION

TEAM ARTEMIS



TEAM MEMBERS:
BESS-MARIE WUDDAH-MARTEY
SANDRA METTLE
GIDEON GLAGO

1. INTRODUCTION

This engineering notebook documents the design, development and testing process of Team Artemis' autonomous robot developed for the World Robot Olympiad (WRO) 2026 Future Engineers Challenge. The purpose of this documentation is to provide a detailed overview of the engineering process followed throughout the creation of the robot, including research, planning, design decisions, prototyping, programming, testing and improvements made through the process.

2. TEAM OVERVIEW

Team Artemis is a team consisting of passionate students from the University of Ghana, Legon. The team was formed with the goal of designing and building an autonomous robot capable of completing the WRO Future Engineers Challenge while applying engineering principles to solve real world problems.

Meet the team:

- a. **Bess-Marie Wuddah-Martey (2nd Year Materials Engineering Student)**
- b. **Sandra Mettle (2nd Year Computer Engineering Student)**
- c. **Gideon Glago (3rd Year Computer Science Student)**

3. RESEARCH AND INSPIRATION

Before designing the robot, extensive research was conducted to understand existing vehicle designs, robotic systems, and engineering concepts relevant to the WRO Future Engineers challenge. This research guided many of the design decisions made throughout the project.

To understand practical vehicle construction, we studied both hobby-grade and toy RC cars, analysing their chassis layouts, steering mechanisms, drivetrain configurations, and overall design. We also reviewed previous WRO Future Engineers solutions to explore different approaches to mechanical design, sensor placement, navigation, and software implementation. This helped us identify effective design practices and potential improvements.

For the mechanical design, we researched Ackermann steering geometry to understand how the inner and outer wheels turn at different angles during cornering, allowing for smoother and more efficient steering. We also investigated differential drive systems to understand how power could be transferred effectively from a single drive motor to both rear wheels while maintaining reliable vehicle performance.

On the software side, we explored sensor fusion techniques to improve localisation and navigation by combining information from multiple sensors. We also evaluated different camera

options for computer vision before selecting a conventional camera that best suited our robot's processing and detection requirements.

Control algorithms were another key area of research. We studied PID control and how adjusting the proportional, integral, and derivative gains affects the robot's steering response and stability. Finally, different motors were compared based on their RPM, torque, and overall performance to ensure the selected motor provided an appropriate balance of speed and driving power.

This research provided the technical foundation for the robot's mechanical, electrical, and software design, allowing subsequent design decisions to be based on established engineering principles rather than trial and error.

4. MECHANICAL DESIGN:

Our primary design objective was to create a compact chassis that could navigate tight turns efficiently while providing enough space to accommodate the battery pack, microcontroller, and other essential electronic components. The chassis was designed in two sections: a lower base to support the drivetrain and motion system, and an upper body to house the remaining electronics.

To minimise weight, the chassis was designed with thin walls while reinforcing critical load-bearing areas to maintain sufficient strength. During the modelling process, iterations were made to improve steering clearance by introducing cut-outs around the front wheel area, ensuring the wheels could achieve larger steering angles without contacting the chassis

Following assembly and testing, additional mounting holes, reinforcement features, and modular slots were incorporated to simplify assembly and allow future modifications or replacement upper-body designs without changing the base chassis.

4.1. STEERING SYSTEM

To satisfy the competition requirements, the robot was designed using an Ackermann steering system. The steering geometry was modelled and verified in CAD to ensure the front wheels followed the correct steering relationship during cornering, reducing tyre scrubbing and improving turning performance.

Most of the optimisation was completed during the CAD stage. Motion simulations were used to identify areas where the wheels interfered with the chassis, allowing the necessary cut-outs and clearances to be incorporated before fabrication. As a result, only minor adjustments were required after the chassis was manufactured.

4.2. DRIVE SYSTEM

The drivetrain was built around an EV3 differential gearbox, allowing power from a single drive motor to be transmitted to both rear wheels while maintaining smooth vehicle motion. During the initial design, a **2:1 gear ratio** was selected to increase the wheel speed, as the chosen motor had a rated speed of **500 RPM**.

However, testing showed that although the higher gear ratio produced greater wheel speed, the available torque was insufficient for reliable starting and acceleration. To address this, the drivetrain was redesigned with a **1:1 gear ratio**, providing greater torque while maintaining adequate speed for the competition. This configuration produced more consistent performance and was therefore adopted as the final drivetrain design.

MODELS

5. ELECTRONICS

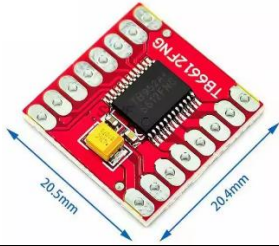
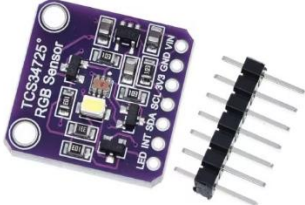
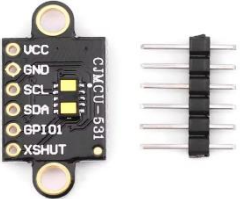
The robot's power system was designed around a rechargeable lithium-ion battery pack consisting of three 18650 cells connected in series. This configuration provided a nominal voltage of approximately **11.1 V**, which was selected as a suitable power source for the 12 V rated drive motor while maintaining a lightweight and compact design.

A voltage regulation system was incorporated to provide appropriate power levels for the different electronic components. Since the battery voltage was higher than the required operating voltage of the control electronics, a buck converter was used to step down the voltage to a suitable level for the Raspberry Pi and other low-power components.

Additional electronic components, including resistors and capacitors, were incorporated into the circuit to support stable operation, reduce electrical noise, and provide protection for sensitive components. This ensured that fluctuations from the motor system did not negatively affect the performance of the control system.

In addition, the components were connected using jumper wires, with thicker gauge wires used for higher current paths to reduce the risk of overheating and ensure reliable power delivery.

Find below a table of components and their uses:

COMPONENTS	QUANTITY	IMAGE	BRIEF USAGE DESCRIPTION
Raspberry Pi 3B+	1		microcontroller/microprocessor
OV5647 Camera 160° wide angle	1		Camera for obstacle detection
TB6612FNG Motor Driver	1		Motor Driver
TCS34725 Color Sensor	1		Colour sensor to detect coloured lines
VL53L1X ToF sensor	4		Distance detector for walls

10 DoF IMU Sesnor	1		Calculating/Measuring accurate angles of turn
Rechargeable Batteries (pack/ singular with battery holder) 12V	1		Power Supply
XL4015 Step-down back convertor	1		Voltage regulation
Switch	1		Power key
Push button	1		Start and stop button

6. SOFTWARE DESIGN AND ARCHITECTURE

The software system was developed with the goal of creating a reliable and adaptable control framework that could be tested extensively before deployment on the physical robot. A key design decision was to separate the robot's decision-making logic from the hardware-specific components, allowing the same control algorithms used in simulation to be transferred directly to the real robot without major modifications.

The software architecture was divided into three main layers: the **core control system**, the **simulation environment**, and the **hardware interface layer**. The core layer contains the robot's main logic, including sensor processing, navigation algorithms, and control strategies. The simulation layer was developed to model the WRO environment and robot behaviour, allowing different strategies to be tested and improved before implementation. The hardware layer manages communication with the physical components, including sensors, motors, steering mechanisms, and the Raspberry Pi.

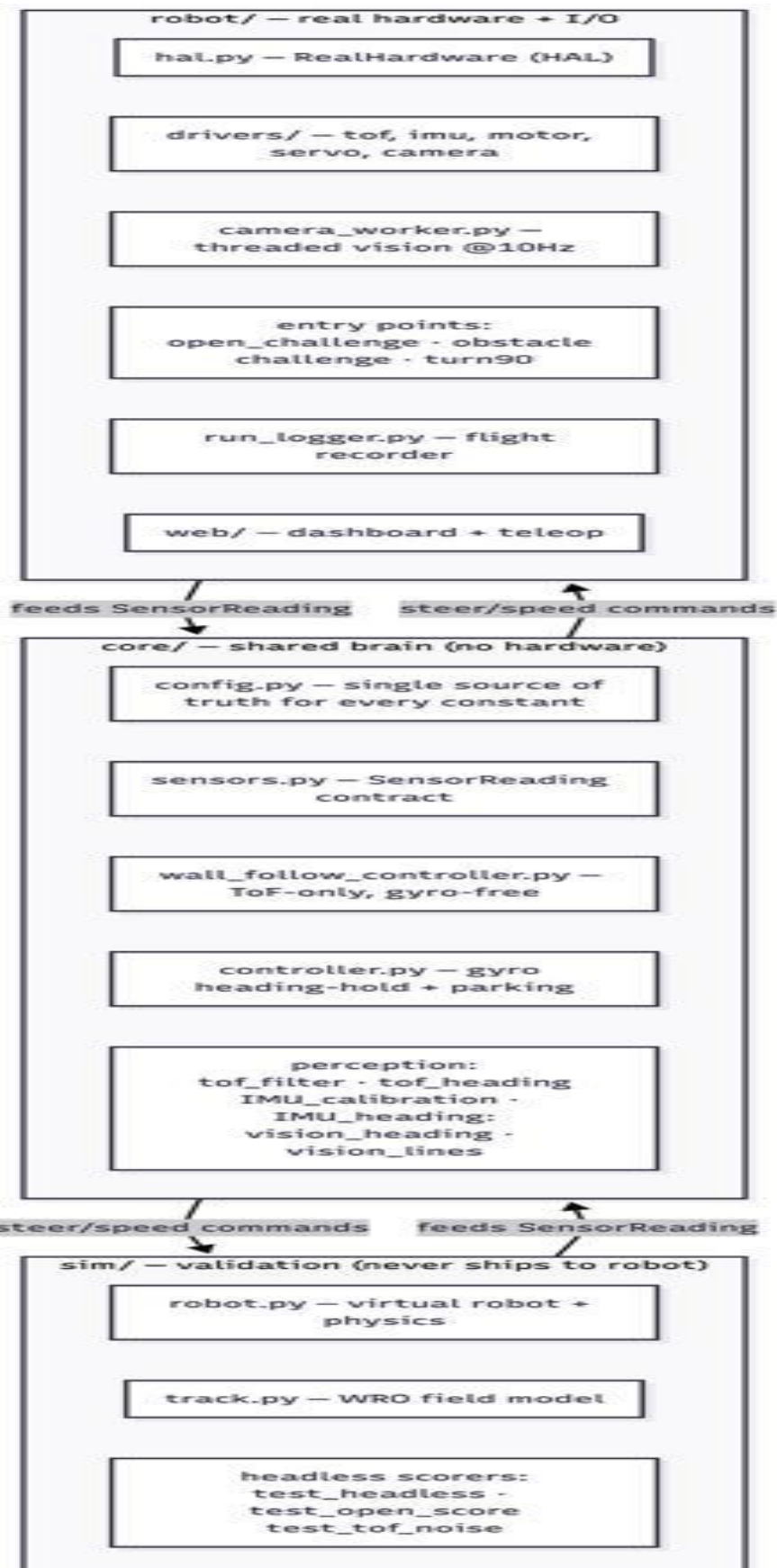
A major focus of the software development was sensor integration. Data from different sensors, including Time-of-Flight (ToF) sensors, IMU, and camera-based vision, was processed and combined to improve the robot's understanding of its environment. This approach, known as sensor fusion, allowed the robot to make more reliable navigation decisions by reducing dependence on a single sensor source.

The robot was designed with multiple navigation strategies to improve reliability. A ToF-based wall-following approach was developed as a simple and robust fallback method, while more advanced modes incorporated camera and IMU data for improved heading estimation and corner detection. PID control techniques were implemented to regulate movement and steering by continuously adjusting the robot's response based on the difference between its desired and actual behaviour.

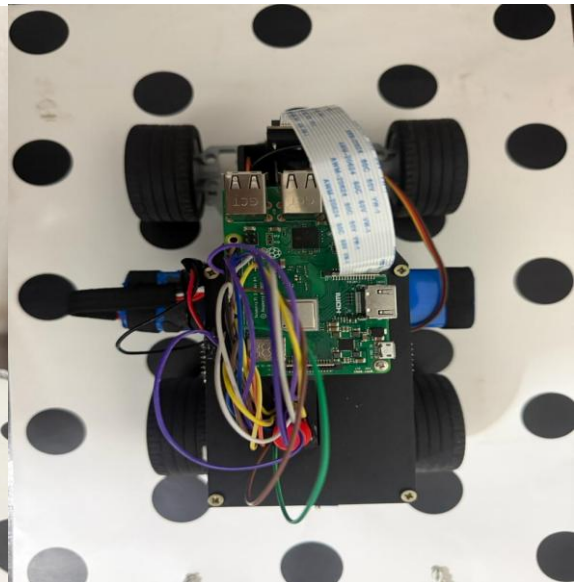
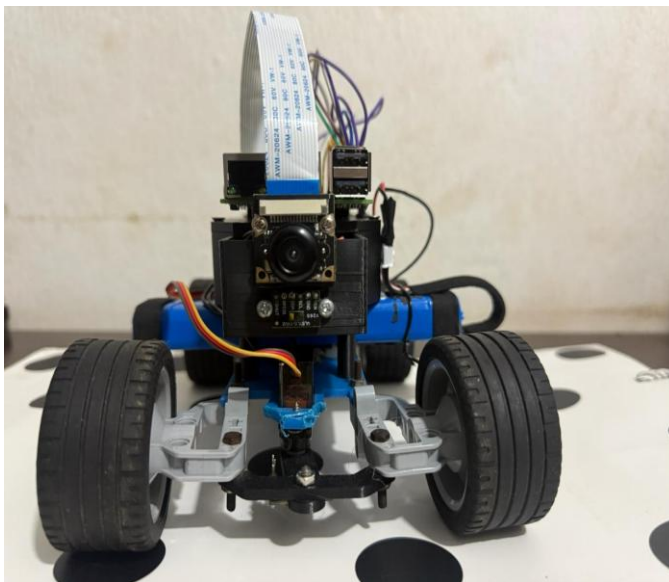
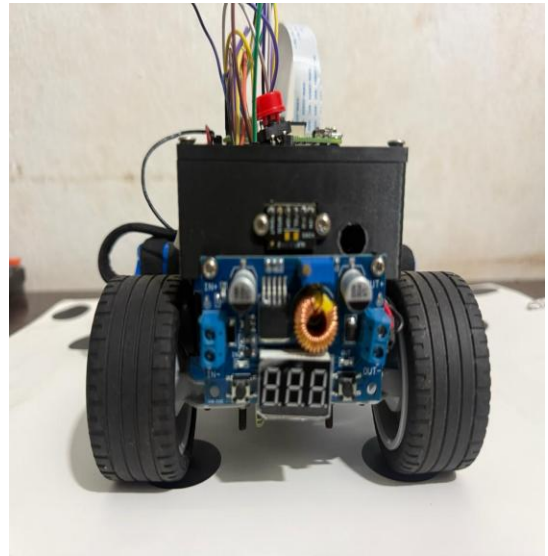
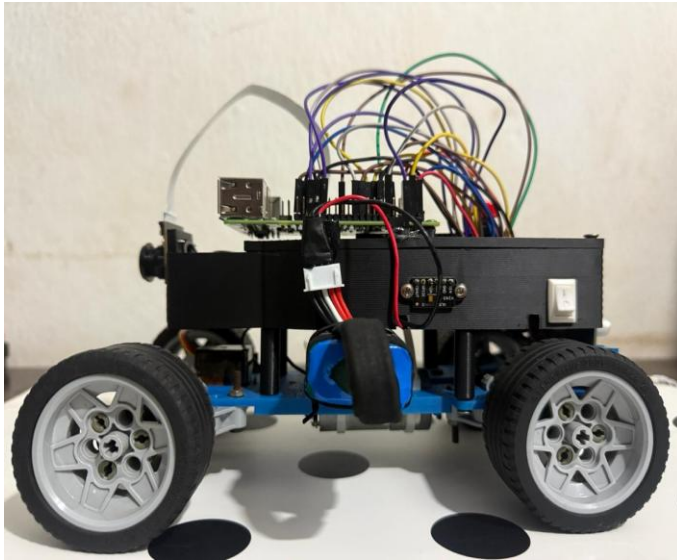
Before testing on the physical robot, the control algorithms were validated using a custom simulation environment. The simulator allowed the team to evaluate performance, introduce variations in conditions, and identify weaknesses in the control system without risking damage to the robot. After validation, the same software framework was deployed on the Raspberry Pi, where it controlled the actual robot through a hardware abstraction layer.

To improve debugging and performance analysis, a logging system was incorporated to record sensor readings, camera detections, and robot behaviour during testing. This data allowed the team to analyse failures, tune parameters, and make informed improvements during development.

BLOCK CHART



7. GALLERY



8. CONCLUSION

The development of Team Artemis' robot involved the integration of mechanical design, electronics, and software engineering principles to create a reliable autonomous vehicle for the WRO Future Engineers challenge. Throughout the design process, emphasis was placed on creating a compact and efficient system capable of navigating the competition environment while maintaining stability, accuracy, and adaptability.

The mechanical design focused on achieving effective movement through a compact chassis, Ackermann steering system, and a differential drivetrain. Multiple design iterations were carried out to improve steering clearance, structural strength, and overall performance. The electronic

system was developed to provide reliable power distribution and communication between the Raspberry Pi, sensors, motor driver, and other components, while the software architecture focused on creating a robust control system supported by simulation, sensor processing, and feedback-based control.

A major aspect of the project was the use of iterative design and testing. Simulations, CAD analysis, and physical testing were used to identify limitations and guide improvements throughout development. By separating the software control logic from the hardware implementation, the robot was able to be tested and refined efficiently before deployment.

Overall, the project provided valuable experience in applying engineering concepts to a real-world autonomous system. Through continuous improvement and integration of mechanical, electronic, and software solutions, Team Artemis developed a robot that reflects a balance between performance, reliability, and practical design considerations.